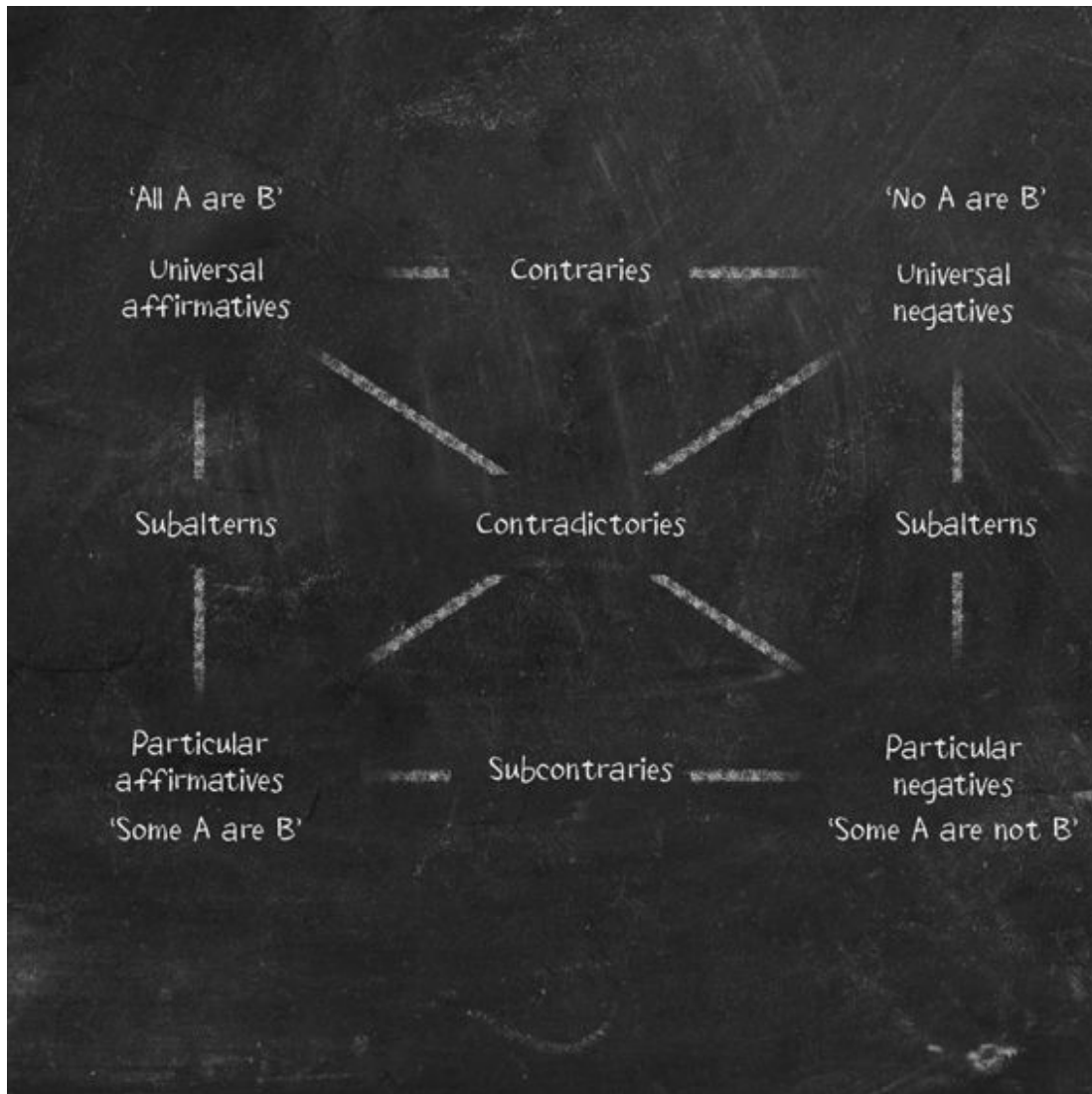


Logic and the syllogism

Like all philosophers, Aristotle sought to justify his theories with rational arguments. But he was not content with the simple reasoning used by the early philosophers, nor even the dialectical method developed by Socrates. Instead, he proposed a system of logic whereby the information in two statements or 'premises' can be used to reach a conclusion. For example, from the premises 'All men are mortal' and 'Socrates is a man', we can infer the conclusion 'Socrates is mortal'.

In the first formal study of logic, Aristotle broke down this form of logical argument, known as a syllogism, into three parts: a major premise, a minor premise and a conclusion. Each of these parts contains two terms, presented in various forms such as 'All A are B', 'Some A are B', 'No A are B' or 'Some A are not B'. Using his talent for analysis and classification, he then categorized the possible combinations of the different forms of premise and conclusion, identifying those that presented valid arguments, and those that were invalid.



The square of opposition is a way of representing relationships between the four different types of proposition used in syllogisms.